

Karey Maynor

📍 Atlanta, GA 🔗 karey-maynor.github.io 🆔 0009-0009-3351-8408

Education

- Ph.D. The University of Texas at Austin**, Mechanical Engineering Austin, TX
 • Dissertation: *Techno-economic enhancement of hydrate-based technologies and direct air capture* Aug 2021 – Aug 2026
 • Advisor: Dr. Vaibhav Bahadur
- B.S. Texas A&M University**, Chemical Engineering College Station, TX
 • Safety Engineering Certificate; Minors in Business and Chemistry May 2017

Research Experience

- Georgia Institute of Technology**, Postdoctoral Fellow, Water-Energy Research Laboratory (WERL) Atlanta, GA
 Desalination systems for water production and brine concentration. Sept 2026 – present
- Bahadur Research Group, The University of Texas at Austin**, NSF Graduate Research Fellow Austin, TX
 Sept 2023 – Aug 2026
- Conducted CO₂ hydrate formation experiments with CO₂/N₂ gas mixtures in a bubble column reactor to evaluate a magnesium kinetic promoter and the effect of lower-concentration CO₂ streams on formation rates.
 - Modeled the techno-economics of hydrate-based desalination from mixed gas sources, benchmarked against pure CO₂ sequestration and leading desalination technologies.
 - Developed a life cycle analysis of CO₂ separation and seabed sequestration via hydrates.
 - Designed thermal testbeds and ANSYS CFD models to quantify heat transfer enhancement in heat pipes and interposers for next-generation electronic packages (two Texas Institute for Electronics projects).
 - Collaborated with Instituto Superior Técnico, University of Lisbon, on magnesium-based ionic liquids for CCUS and gas separation.
 - Developed first-principles, system-level, and techno-economic models of a novel mass exchanger for direct air capture.
 - Assessed the cost and emissions impact of electrolytic hydrogen on Midwest ethanol using Aspen Plus, H2A-Lite, and GREET.
- Bahadur Research Group, The University of Texas at Austin**, Graduate Research Assistant Austin, TX
 Mar 2022 – Sept 2023
- Designed and executed THF hydrate formation studies to understand magnesium-based nucleation promotion.
 - Built an Aspen Plus process model and MATLAB techno-economic model of clean hydrogen-based ammonia production.
 - ARPA-E project: developed insulating materials to extend power transformer life; used experimental data to refine an ANSYS thermal model and characterized thermal degradation via viscosity and tensile strength.
- The University of Texas at Austin**, Graduate Teaching Assistant, ME 139L Experimental Heat Transfer Lab Austin, TX
 Aug 2022 – Apr 2023
- Led two lab sections per semester covering experimental design, uncertainty analysis, and heat transfer systems analysis.

Systems Optimization & Multi-Scale Analysis Lab, Texas A&M University, Undergraduate Research Assistant
• Investigated water extraction and purification from effluent streams to develop a process superstructure and MINLP model.

College Station, TX
June 2016 – Aug 2016

Professional Experience

Heat Transfer Research, Inc. (HTRI), Test Engineer I & II

Navasota, TX

- Ran research experiments on closed-loop pilot-scale systems studying heat transfer and fluid dynamics.
- Designed and directed a \$107k unit upgrade, delivered 33% under budget and two months ahead of schedule.
- Technical Lead, ISO 17025 quality management: led the company through two reaccreditation audits and managed technical staff on certified testing contracts.
- Safety Program Lead: led a team of management and technicians to overhaul safety practices and standardized HTRI's HAZOP study process.

June 2017 – June 2021

Honors and Awards

NSF Graduate Research Fellowship
National Science Foundation

2023

Leigh Family Endowed Graduate Fellowship in Mechanical Engineering
The University of Texas at Austin

Aug 2025

Professional Development Award (2023, 2025)
UT Austin Graduate School

Aug 2025

Chevron Energy Graduate Fellow Award
Chevron

Sept 2023

Cockrell School of Engineering Thrust 2000 Fellowship (2021–2025)
Cockrell School of Engineering

Aug 2021

T.W. Whaley, Jr. Friends of Alec Endowed Scholarship (2021–2022)
The University of Texas at Austin

Aug 2021

Publications

Ultrafast Formation of Carbon Dioxide Hydrates from Mixtures of Carbon Dioxide and Nitrogen Gases

2026

K. Maynor, M. Hamalian, V. Bahadur

[10.1016/j.ijggc.2026.104751](https://doi.org/10.1016/j.ijggc.2026.104751) (International Journal of Greenhouse Gas Control, 155, 104751)

Review of Clathrate Hydrate-Based CO₂ Capture: Benchmarking Performance, Economics, and Life Cycle Considerations

2026

K. Maynor, M. Hamalian, S. Srivastava, J. S. Pandey, V. Bahadur

[10.1016/j.jece.2026.124335](https://doi.org/10.1016/j.jece.2026.124335) (Journal of Environmental Chemical Engineering, 14(6), 124335)

Levelized Cost Pass-through of Green Hydrogen in the Bioethanol Fuel Value Chain in U.S. Midwest

2026

K. Maynor, M. Arzumanyan, N. Lin, V. Bahadur

[10.1016/j.seta.2026.104911](https://doi.org/10.1016/j.seta.2026.104911) (Sustainable Energy Technologies and Assessments, 87, 104911)

Influence of Bubble Delivery Parameters on the Formation Kinetics and Composition of Carbon Dioxide Hydrate Slurry

2025

M. Hamalian, **K. Maynor**, V. Bahadur

[10.1016/j.cej.2025.168399](https://doi.org/10.1016/j.cej.2025.168399) (Chemical Engineering Journal, 523, 168399)

- Techno-Economic Analysis of Electrochemical Carbon Capture from Oceanwater Integrated with Hydrates-Based Sequestration** 2025
M. Hamalian, A. Bhati, **K. Maynor**, V. Bahadur
[10.1016/j.apenergy.2025.125960](https://doi.org/10.1016/j.apenergy.2025.125960) (Applied Energy, 392, 125960)
- Carbon Dioxide Hydrate Formation-Based Carbon Capture from Mixtures of Carbon Dioxide and Nitrogen** 2025
K. Maynor, M. Hamalian, V. Bahadur
(Proceedings of the ASME 19th International Conference on Energy Sustainability, Westminster, CO)
- System-Level and Techno-Economic Analysis of Ammonia Production from Clean Hydrogen in the Permian Basin in Texas** 2025
K. Maynor, G. Childers, T. Soori, N. Lin, M. Lewis, V. Bahadur
(Proceedings of the ASME 19th International Conference on Energy Sustainability, Westminster, CO)
- Effects of Bubble Size and Density on CO₂ Hydrate Slurry Production for Carbon Sequestration (Best Paper Award)** 2025
M. Hamalian, **K. Maynor**, V. Bahadur
(Proceedings of the ASME 19th International Conference on Energy Sustainability, Westminster, CO)
- Magnesium-Induced Rapid Nucleation of Tetrahydrofuran Hydrates** 2024
K. Maynor, A. Bhati, M. Hamalian, A. M. Ferraria, A. P. D. C. Ribeiro, A. S. Moita, V. Bahadur
[10.1021/acs.langmuir.4c02882](https://doi.org/10.1021/acs.langmuir.4c02882) (Langmuir, 40(41), 21758–21766)
- Modeling the Impact of High Thermal Conductivity Paper on the Performance and Life of Power Transformers** 2024
S. Bilyaz, A. Bhati, M. Hamalian, **K. Maynor**, T. Soori, A. Gattozzi, C. Penney, D. Weeks, Y. Xu, L. Hu, J. Y. Zhu, J. K. Nelson, R. Hebner, V. Bahadur
[10.1016/j.heliyon.2024.e27783](https://doi.org/10.1016/j.heliyon.2024.e27783) (Heliyon, 10(6), e27783)

Technical Reports and Workshops

- **K. Maynor** & A. P. D. C. Ribeiro. *Carbon Dioxide Hydrates for Carbon Management: Opportunities for Capture, Storage and Utilization*. Workshop, 9th Multidisciplinary Symposium on Circular Economy and Urban Mining, Procida, Italy, 2026.
- M. Arzumanyan, V. Bahadur, L. Ning, T. H. Rupam, **K. Maynor**, S. Katakam, & M. Usama. *Water Use Requirements for Data Centers in Texas* (white paper). Bureau of Economic Geology, The University of Texas at Austin, 2025.
- **K. Maynor** & L. Huang. *PHE-18-TR: Shellside Boiling Tests in Plate-and-Shell Heat Exchanger*. Heat Transfer Research, Inc., 2018.

Presentations

- *Process Intensification to Enable CO₂ Hydrate-based Seabed Carbon Sequestration*. 11th International Conference on Gas Hydrates & 3rd European Conference on Gas Hydrates, Lyon, France, 2026.
- *Process Intensification to Enable CO₂ Hydrate-based Seabed Carbon Sequestration*. Invited seminar, Center for Energy Resource Engineering, Technical University of Denmark, Lyngby, Denmark, 2026.
- *Carbon Dioxide Hydrate Formation-Based Carbon Capture from Mixtures of Carbon Dioxide and Nitrogen* (oral). ASME 19th International Conference on Energy Sustainability, Westminster, CO, 2025.
- *System-Level and Techno-Economic Analysis of Ammonia Production from Clean Hydrogen in the Permian Basin in Texas*. ASME Energy Sustainability, 2025 (oral); UT Hydrogen Workforce Summit, 2024 (poster, 5th of 41); Chevron Energy Fellows Forum, 2024 (oral); H2@Scale in Texas, 2024 (poster); UT Energy Week, 2024 (oral and poster); ASME IMECE, 2023 (oral).

Teaching Certifications

- Future Faculty Fellows Program, Center for Teaching and Learning, Georgia Institute of Technology (in progress)
- Teaching Preparation Series: Advanced Certification, Center for Teaching & Learning, The University of Texas at Austin
- Teaching Assistant Certification, Cockrell School of Engineering, The University of Texas at Austin

Leadership and Mentoring

- **Research Mentor**, Bahadur Research Group, UT Austin (2022–2026): mentored 2 Ph.D., 1 master's, and 2 undergraduate students.
- **President**, Graduate Engineering Council, UT Austin (2021–2024): led the Graduate and Industry Networking Event (GAIN; 60+ poster presenters, 14 industry sponsors); awarded a record \$10,000+ in research and travel grants; served on the Senate of College Councils.
- **Financial Director** and **IT & Webpage Director**, Graduate Engineering Council, UT Austin.
- **Mentor & Mentee**, Mechanical Engineering Mentor Program, UT Austin (2021–2025).

Professional Service

- Session chair, ASME Summer Heat Transfer Conference
- Reviewer: Separation and Purification Technology (journal); ASME Summer Heat Transfer Conference; ASME Energy Sustainability Conference
- Member: ASME, AIChE, American Society of Thermal and Fluids Engineers (ASTFE), International Research Association on Circular Economy (IRACE)

Skills

Programming

Python, MATLAB, LabVIEW

Simulation and Modeling

Aspen Plus, ANSYS (Discovery, Fluent), H2A-Lite, GREET

Research Interests

- Desalination and brine management · Carbon capture, utilization & storage · Gas hydrates · Direct air capture · Thermal and energy systems · Hydrogen · Liquid/gas separations · Techno-economic and life cycle analysis